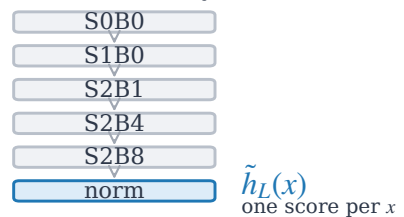
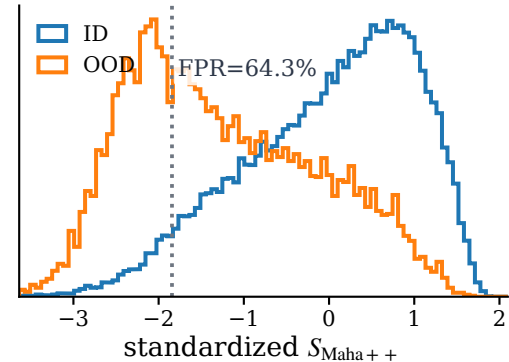


Baseline (Maha++)

Mahalanobis on final-layer features

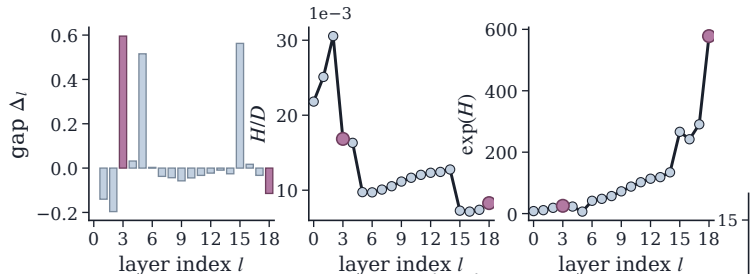


Last-layer Maha++ scores

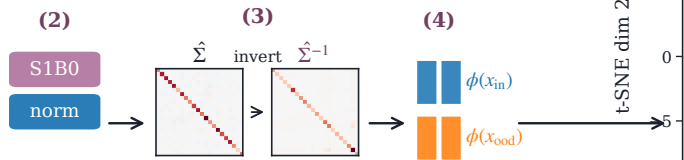


(1) Rank layers

(1a) gap Δ_l · (1b) compression H/D · (1c) effective rank $\exp(H)$

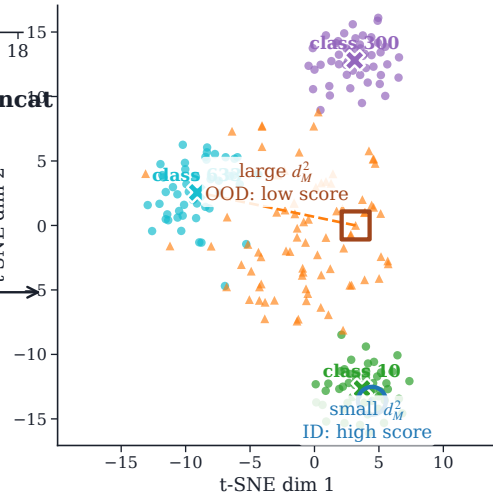


(2) Top-K + ℓ_2 ($K=2$) · (3) $\hat{\Sigma} \rightarrow \hat{\Sigma}^{-1}$ (LW) · (4) Concat

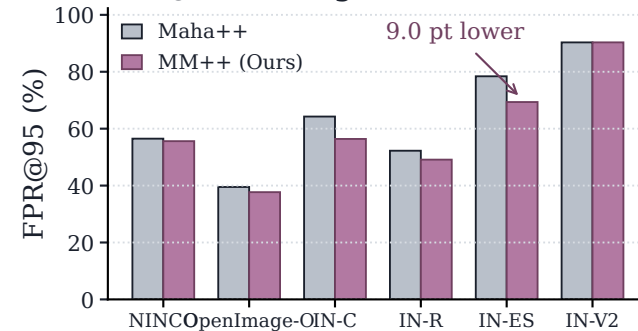


(5) Score

$$S_{\text{MM++}}(x) = -\min_c d_M^2(\phi(x), \hat{\mu}_c^K)$$



FPR@95 on ImageNet-LT OOD benchmarks (lower is better)



MM++ fused scores

